FILTRATION OF A FINITELY GENERATED SUBSHEAF OF THE CONSTANT SHEAF Z BY EXTENSIONS OF EXTENSIONS BY ZERO

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. This card was re-admitted after a gate audit: it had been excluded only because the chapter's retained preamble lacks the reference/slogan/history environments, and the author's own preamble defines them as comment environments. The card therefore carries a per-card preamble copy in its task directory, consisting of the chapter retained preamble copied verbatim plus the project's own three display environments copied verbatim from sources/stacks-chow/preamble.tex lines 7-12. Nothing was added or redefined by the assembler side.

Editorial note. The author's own bibliographic remark, which sits in a reference environment at the head of the statement and cites the original paper, is preserved in the source span and is not typeset because the author's own preamble comments reference environments out; no bibliography excerpt is therefore needed.

Editorial note. The counted spans carry no navigation stub other than those already listed and no bibliography citation reaches the compiled document.

Editorial note. The label lemma-subsheaf-of-constant-sheaf is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Subsheaf generated by local sections. Modules, Definition modules-definition-generated-by-local-sections, and the description of the subsheaf generated by finitely many sections, used throughout proof lines 3504-3526. Supplies the sheaves whose filtration is constructed.
- Extension by zero along the inclusion of an open subspace. Sheaves, the extension by zero functor named in the statement at lines 3496-3500. Supplies the terms of the short exact sequences.
- Sections of the constant sheaf Z are locally constant integer valued functions. Standard, used at proof lines 3505-3508. Supplies the decomposition that makes the generators constant.

Lemma 0.1. *Let X be a topological space such that the intersection of any two quasi-compact opens is quasi-compact. Let $\mathcal{F} \subset \underline{\mathbf{Z}}$ be a subsheaf generated by finitely many sections over quasi-compact opens. Then there exists a finite filtration*

$$(0) = \mathcal{F}_0 \subset \mathcal{F}_1 \subset \ldots \subset \mathcal{F}_n = \mathcal{F}$$

by abelian subsheaves such that for each $0 < i \leq n$ there exists a short exact sequence

$$0 \to j'_!\underline{\mathbf{Z}}_V \to j_!\underline{\mathbf{Z}}_U \to \mathcal{F}_i/\mathcal{F}_{i-1} \to 0$$

with $j : U \to X$ and $j' : V \to X$ the inclusion of quasi-compact opens into X.

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Proof. Say $\mathcal{F}$ is generated by the sections $s_1, \ldots, s_t$ over the quasi-compact opens $U_1, \ldots, U_t$. Since U_i is quasi-compact and s_i a locally constant function to $\mathbf{Z}$ we may assume, after possibly replacing U_i by the parts of a finite decomposition into open and closed subsets, that s_i is a constant section. Say $s_i = n_i$ with $n_i \in \mathbf{Z}$. Of course we can remove (U_i, n_i) from the list if $n_i = 0$. Flipping signs if necessary we may also assume $n_i > 0$. Next, for any subset $I \subset \{1, \ldots, t\}$ we may add $\bigcap_{i \in I} U_i$ and $\gcd(n_i, i \in I)$ to the list. After doing this we see that our list $(U_1, n_1), \ldots, (U_t, n_t)$ satisfies the following property: For $x \in X$ set $I_x = \{i \in \{1, \ldots, t\} \mid x \in U_i\}$. Then $\gcd(n_i, i \in I_x)$ is attained by n_i for some $i \in I_x$.

As our filtration we take $\mathcal{F}_0 = (0)$ and $\mathcal{F}_n$ generated by the sections n_i over U_i for those i such that $n_i \leq n$. It is clear that $\mathcal{F}_n = \mathcal{F}$ for $n \gg 0$. Moreover, the quotient $\mathcal{F}_n/\mathcal{F}_{n-1}$ is generated by the section n over $U = \bigcup_{n_i \leq n} U_i$ and the kernel of the map $j_!\underline{\mathbf{Z}}_U \to \mathcal{F}_n/\mathcal{F}_{n-1}$ is generated by the section n over $V = \bigcup_{n_i \leq n-1} U_i$. Thus a short exact sequence as in the statement of the lemma. $\square$